Respiratory System

- Anatomy
  - Upper respiratory tract
    - Nose
    - Nasal cavity
    - Pharynx
    - Larynx
  - Lower respiratory tract
    - Trachea
    - Bronchi & Bronchioles
    - Lungs
    - Alveoli
  - Supporting structures
    - Diaphragm
    - Intercostal muscles
    - Pleura
- Medical Conditions
  - Asthma
  - COPD
  - Pneumonia
  - Tuberculosis
  - Pulmonary fibrosis
  - Pulmonary embolism
  - ARDS
  - Sleep apnea
- Types of Respiratory Diseases
  - Obstructive (asthma, COPD)
  - Restrictive (fibrosis, scoliosis)
  - Infectious (TB, pneumonia, COVID-19)
  - Environmental/Occupational (silicosis, asbestosis)
  - Vascular (pulmonary embolism)
- Symptoms
  - Cough (dry/wet)
  - Breathlessness
  - Wheezing
  - Chest pain
  - Fatigue
  - Fever
  - Sputum production
  - Cyanosis
- Causes
  - Infections (bacterial/viral)
  - Smoking
  - Air pollution
  - Occupational exposure
  - Allergens
  - Genetics
  - Sedentary lifestyle
  - Immune issues
- Diagnosis
  - Physical exam
  - Chest X-ray
  - CT scan
  - Spirometry
  - ABG analysis
  - Pulse oximetry
  - Blood tests
  - Bronchoscopy
- Treatment
  - Medications
    - Bronchodilators
    - Steroids
    - Antibiotics
    - Antivirals
  - Therapies
    - Oxygen therapy
    - Pulmonary rehab
    - Inhalation therapy
  - Surgical
    - Lung transplant
    - Thoracic surgery
- Prognosis
  - Depends on disease type
  - Early diagnosis → better outcomes
  - Chronic diseases: managed, not cured
  - Impact of lifestyle (smoking, pollution)
- Rehabilitation
  - Breathing exercises
  - Spirometry training
  - Yoga & physiotherapy
  - Lifestyle modification
  - Nutrition
  - Long-term monitoring
- Devices (Include Startups)
  - Clinical Devices
    - Ventilators
    - CPAP/BiPAP
    - Pulse oximeters
    - Nebulizers
    - Spirometers
    - Oxygen concentrators
  - Innovative / Startup Devices
    - AI-based portable spirometers (AsthmaAI, AlveoAir)
    - Smart inhalers (Propeller Health)
    - Lung imaging wearables
    - Home-based respiratory monitoring IoT devices
    - Low-cost oxygen concentrators (startups in India)
- Competitive Landscape
  - Philips Healthcare
  - Medtronic
  - ResMed
  - GE Healthcare
  - Dräger
  - Indian players
    - AgVa Healthcare
    - Phoenix Medical Systems
    - AlveoAir
    - Noccarc
- Market Study
  - Market growth drivers
    - Pollution
    - Smoking
    - Aging population
    - COVID-19 impact
  - Global respiratory device market > $25B+
  - High growth areas
    - Ventilators
    - Home oxygen therapy
    - Digital respiratory monitoring
  - India: Rapid adoption of affordable devices
- Problems Faced
  - Underdiagnosis of diseases
  - High cost of advanced devices
  - Poor air quality in cities
  - Low access in rural regions
  - Compliance issues with inhalers
  - Limited awareness of pulmonary rehab
  - Shortage of pulmonologists